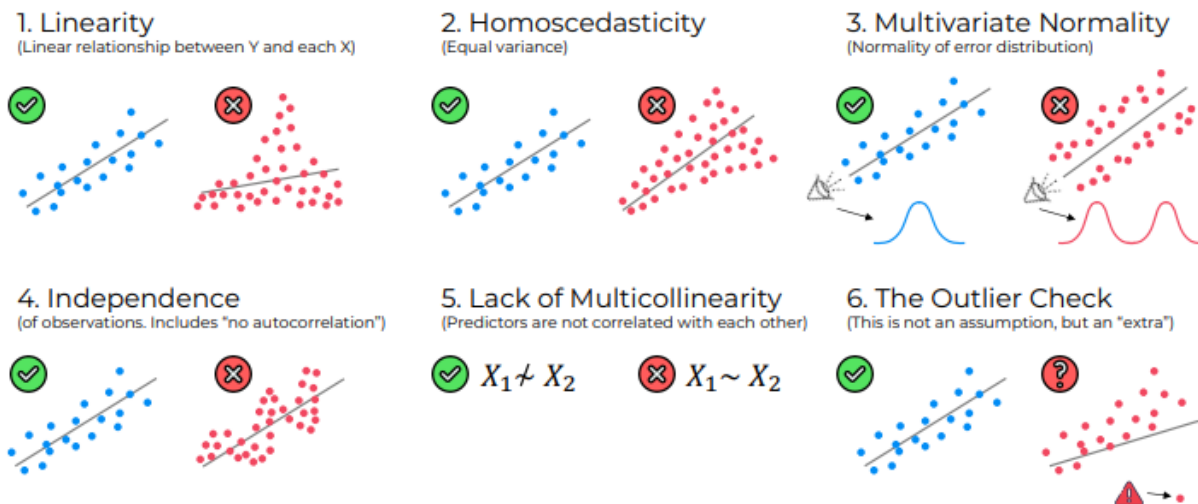


Machine Learning (CAI-262)

Objectives:

- Assumptions of Linear Regression
- Performance Measures for Regression
- Correlation

Assumptions of Linear Regression



Assessing the performance of Regression model:

Assessing the performance of a linear regression model involves evaluating how well the model fits the data and makes predictions. Several metrics can be used to evaluate the performance of regression models, some of them are given below.

1. Mean Absolute Error (MAE):

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|$$

MAE is a very simple metric which calculates the absolute difference between actual and predicted values. To better understand, let's take an example you have input data and output data and use Linear Regression, which draws a best-fit line.

Now you have to find the MAE of your model which is basically a mistake made by the model known as an error. Now find the difference between the actual value and predicted value that is an absolute error but we have to find the mean absolute of the complete dataset.

so, sum all the errors and divide them by a total number of observations And this is MAE. And we aim to get a minimum MAE because this is a **loss**.

2. Mean Absolute Percentage Error (MAPE):

$$MAPE = \frac{1}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

3. Mean Squared Error (MSE):

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

MSE is a most used and very simple metric with a little bit of change in mean absolute error. Mean squared error states that finding the squared difference between actual and predicted value. So, above we are finding the absolute difference and here we are finding the squared difference. What actually the MSE represents? It represents the squared distance between actual and predicted values. we perform squared to avoid the cancellation of negative terms and it is the benefit of MSE.

2. Root Mean Squared Error (RMSE):

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}$$

As RMSE is clear by the name itself, that it is a simple square root of mean squared error.

3. R-squared (R2):

$$R^2 = 1 - \frac{RSS}{TSS} = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}$$

R2 score is a metric that tells the performance of your model, not the loss in an absolute sense that how many wells did your model perform.

In contrast, MAE and MSE depend on the context as we have seen whereas the R2 score is independent of context. So, with help of R squared we have a baseline model to compare a model which none of the other metrics provides. The same we have in classification problems

which we call a threshold which is fixed at 0.5. So basically R2 squared calculates how much regression line is better than a mean line.

Hence, R2 squared is also known as **Coefficient of Determination** or sometimes also known as Goodness of fit. If the **R2 score is 0** then the above regression line by mean line is equal means 1 so 1-1 is zero. So, in this case, both lines are overlapping means model performance is worst, It is not capable to take advantage of the output column.

Now the second case is when the **R2 score is 1**, it means when the division term is zero and it will happen when the regression line does not make any mistake, it is perfect. In the real world, it is not possible. So we can conclude that as our regression line moves towards perfection, R2 score move towards one. And the model performance improves.

The normal case is when the **R2 score is between zero and one** like 0.8 which means your model is capable to explain 80 per cent of the variance of data.

4. AVERAGE R-SQUARED (Average R2)

- Average R-squared (Average R2) is used when we want to evaluate the performance of a regression model across multiple subsets of data.
- It calculates the mean of the R-squared values obtained from applying the model to each subset of data.
- Example: Suppose we want to assess the performance of our house price prediction model in different regions of the country. We divide our dataset into subsets based on regions (e.g., North, South, East, West). We then build a separate regression model for each subset and calculate the R-squared value for each model. Average R2 computes the mean of these R-squared values to provide an overall measure of model performance across all regions.

Correlation:

Correlation can be defined as a measurement that is used to quantify the relationship between variables. If an increase (or decrease) in one variable causes a corresponding increase (or decrease) in another then the two variables are said to be **directly correlated**. Similarly, if an

increase in one causes a decrease in another or vice versa, then the variables are said to be **indirectly correlated**. If a change in an independent variable does not cause a change in the dependent variable then they are **uncorrelated**. Thus, correlation can be positive (direct correlation), negative (indirect correlation), or zero. This relationship is given by the correlation coefficient.

The best way to conduct correlation and regression analysis is by using **Pearson's correlation coefficient** and by adopting the method of **least squares** respectively. The correlation formula is given below:

$$\text{Pearson's Correlation Coefficient: } r_{xy} = \frac{\sum_1^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_1^n (x_i - \bar{x})^2 \sum_1^n (y_i - \bar{y})^2}}$$

Difference between Correlation and Regression

Correlation and regression are both used as statistical measurements to get a good understanding of the relationship between variables. If the correlation coefficient is negative (or positive) then the slope of the regression line will also be negative (or positive). The table given below highlights the key difference between correlation and regression.

Correlation	Regression
Correlation is used to determine whether variables are related or not.	Regression is used to numerically describe how a dependent variable changes with a change in an independent variable
Correlation tries to establish a linear relationship between variables.	It finds the best-fitted regression line to estimate an unknown variable on the basis of the known variable.
The variables can be used interchangeably	The variables cannot be interchanged.
Correlation uses a signed numerical value to estimate the strength of the relationship between the variables.	Regression is used to show the impact of a unit change in the independent variable on the dependent variable.
The Pearson's coefficient is the best measure of correlation.	The least-squares method is the best technique to determine the regression line.

Example 1: Calculate the correlation coefficient for the given data

Person	Hand length	Height
A	17	150
B	15	154
C	19	169
D	17	172
E	21	175

Solution:

Person	Hand	Height	$(x_i - \bar{x})$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$
A	17	150	-0.8	-14.0	11.2	0.6	196.0
B	15	154	-2.8	-10.0	28.0	7.8	100.0
C	19	169	1.2	5.0	6.0	1.4	25.0
D	17	172	-0.8	8.0	-6.4	0.6	64.0
E	21	175	3.2	11.0	35.2	10.2	121.0
Average	17.8	164		Total	74.0	20.8	506.0

- Using the formula,
- $r_{xy} = 0.72$
- **Answer:** The data has a high positive correlation

Example 2: Interpret the correlation coefficient of the following data

X	Y
12	1002
14	760
21	580
34	400
56	350
78	120

Solution:

X	Y	$(x_i - \bar{x})$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$
12	1002	-23.833	466.67	-11122.22	568.028	217777.78
14	760	-21.833	224.67	-4905.22	476.694	50475.11
21	580	-14.833	44.67	-662.56	220.028	1995.11
34	400	-1.833	-135.33	248.11	3.361	18315.11
56	350	20.167	-185.33	-3737.56	406.694	34348.44
78	120	42.167	-415.33	-17513.22	1778.028	172501.78
$\bar{x} = 35.83$	$\bar{y} = 535.83$		Total	-37692.667	3452.833	495413.33

- Using the formula,
- $r_{xy} = -0.9114$
- **Answer:** There is a strong negative correlation between the two variables